

Q1 (75)

$$\begin{aligned}
 \vec{C} &= \lambda(\vec{b} \times (\vec{a} \times \vec{b})) \\
 &= \lambda((\vec{b} \cdot \vec{b})\vec{a} - (\vec{b} \cdot \vec{a})\vec{b}) \\
 &= \lambda(5(-\hat{i} + \hat{j} + \hat{k}) + 2\hat{i} + \hat{k}) \\
 &= \lambda(-3\hat{i} + 5\hat{j} + 6\hat{k}) \\
 \vec{C} \cdot \vec{a} &= 7 \Rightarrow 3\lambda + 5\lambda + 6\lambda = 7 \\
 \lambda &= \frac{1}{2} \\
 \therefore 2 \left| \left(\frac{-3}{2} - 1 + 2 \right) \hat{i} + \left(\frac{5}{2} + 1 \right) \hat{j} + (3 + 1 + 1) \hat{k} \right| \\
 &= 2 \left(\frac{1}{4} + \frac{49}{4} + 25 \right) = 25 + 50 = 75
 \end{aligned}$$

Q2 (12)

$$\begin{aligned}
 \vec{r} \times \vec{a} &= \vec{c} \times \vec{a} \\
 \vec{r} \times \vec{a} - \vec{c} \times \vec{a} &= 0 \\
 (\vec{r} - \vec{c}) \times \vec{a} &= 0 \\
 \therefore \vec{r} - \vec{c} &= \lambda \vec{a} \\
 \vec{r} &= \lambda \vec{a} + \vec{c} \\
 \vec{r} \cdot \vec{b} &= \lambda \vec{a} \cdot \vec{b} + \vec{c} \cdot \vec{b} = 0 \\
 \Rightarrow \lambda(1 - 2) + 2 &= 0 \\
 \Rightarrow \lambda &= 2 \\
 \therefore \vec{r} &= 2\vec{a} + \vec{c} \\
 \vec{r} \cdot \vec{a} &= 2|\vec{a}|^2 + \vec{a} \cdot \vec{c} \\
 &= 2(1 + 4 + 1) + (1 - 2 + 1) \\
 &= 12
 \end{aligned}$$

Q3 (2)

$$\begin{aligned}
 \vec{a} &= \hat{i} + \alpha \hat{j} + 3\hat{k} \\
 \vec{b} &= 3\hat{i} - \alpha \hat{j} + \hat{k} \\
 \text{Area of parallelogram} &= |\vec{a} \times \vec{b}|
 \end{aligned}$$

Hints and Solutions

MathonGo

$$= |(\hat{i} + \alpha\hat{j} + 3\hat{k}) \times (3\hat{i} - \alpha\hat{j} + \hat{k})|$$

$$8\sqrt{3} = |(4\alpha)\hat{i} + 8\hat{j} - (4\alpha)\hat{k}|$$

$$(64)(3) = 16\alpha^2 + 64 + 16\alpha^2$$

$$(64)(3) = 32\alpha^2 + 64$$

$$6 = \alpha^2 + 2$$

$$\alpha^2 = 4$$

$$\therefore \vec{a} = \hat{i} + \alpha\hat{j} + 3\hat{k}$$

$$\vec{b} = 3\hat{i} - \alpha\hat{j} + \hat{k}$$

$$\vec{a} \cdot \vec{b} = 3 - \alpha^2 + 3$$

$$= 6 - \alpha^2$$

$$= 6 - 4$$

$$= 2$$

Q4 (3)

$$\vec{a} \times \left(\vec{a} \times \left((\vec{a} \cdot \vec{b})\vec{a} - |\vec{a}|^2\vec{b} \right) \right)$$

$$\vec{a} \times \left(-|\vec{a}|^2(\vec{a} \times \vec{b}) \right) = -|\vec{a}|^2 \left((\vec{a} \cdot \vec{b})\vec{a} - |\vec{a}|^2\vec{b} \right)$$

$$= -(\vec{a} \cdot \vec{b})\vec{a}|\vec{a}|^2 + |\vec{a}|^4\vec{b}$$

$$= |\vec{a}|^4\vec{b} (\because \vec{a} \cdot \vec{b} = 0)$$

Q5 (3)

$$\frac{x}{1} = -\frac{1}{y} = \frac{1}{z} = \lambda (\text{let})$$

$$\text{Unit vector parallel to } x\hat{i} + y\hat{j} + z\hat{k} = \pm \frac{(\lambda\hat{i} - \frac{1}{\lambda}\hat{j} + \frac{1}{\lambda}\hat{k})}{\sqrt{\lambda^2 + \frac{2}{\lambda^2}}}$$

$$\text{For } \lambda = 1, \text{ it is } \pm \frac{(\hat{i} - \hat{j} + \hat{k})}{\sqrt{3}}$$

Q1: 24 Feb (Shift 1) - Numerical

Let three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ be such that $\vec{c}$ is coplanar with $\vec{a}$ and $\vec{b}$, $\vec{a} \cdot \vec{c} = 7$ and $\vec{b}$ is perpendicular to $\vec{c}$, where $\vec{a} = -\hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} + \hat{k}$, then the value of $2|\vec{a} + \vec{b} + \vec{c}|^2$ is

Q2: 25 Feb (Shift 1) - Numerical

Let $\vec{a} = \hat{i} + 2\hat{j} - \hat{k}$, $\vec{b} = \hat{i} - \hat{j}$ and $\vec{c} = \hat{i} - \hat{j} - \hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{a} = \vec{c} \times \vec{a}$ and $\vec{r} \cdot \vec{b} = 0$, then $\vec{r} \cdot \vec{a}$ is equal to

Q3: 25 Feb (Shift 2) - Numerical

Let $\vec{a} = \hat{i} + \alpha\hat{j} + 3\hat{k}$ and $\vec{b} = 3\hat{i} - \alpha\hat{j} + \hat{k}$. If the area of the parallelogram whose adjacent sides are represented by the vectors $\vec{a}$ and $\vec{b}$ is $8\sqrt{3}$ square units, then $\vec{a} \cdot \vec{b}$ is equal to

Q4: 26 Feb (Shift 1) - Single Correct

If $\vec{a}$ and $\vec{b}$ are perpendicular vectors, then

$\vec{a} \times (\vec{a} \times (\vec{a} \times (\vec{a} \times \vec{b})))$ is equal to

(1) $\frac{1}{2}|\vec{a}|^4\vec{b}$

(2) $\vec{a} \times \vec{b}$

(3) $|\vec{a}|^4\vec{b}$

(4) 0

Q5: 26 Feb (Shift 2) - Single Correct

If vectors $\vec{a}_1 = x\hat{i} - \hat{j} + \hat{k}$ and $\vec{a}_2 = \hat{i} + y\hat{j} + z\hat{k}$ are collinear, then a possible unit vector parallel to the vector $x\hat{i} + y\hat{j} + z\hat{k}$ is :

(1) $\frac{1}{\sqrt{2}}(-\hat{j} + \hat{k})$

(2) $\frac{1}{\sqrt{2}}(\hat{i} - \hat{j})$

(3) $\frac{1}{\sqrt{3}}(\hat{i} - \hat{j} + \hat{k})$

Questions with Answer Keys

MathonGo

(4) $\frac{1}{\sqrt{3}}(\hat{i} + \hat{j} - \hat{k})$

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

Answer Key**Q1 (75)****Q2 (12)****Q3 (2)****Q4 (3)****Q5 (3)**